Descriptive Analytics Report: Flipkart

Product Catalog Behavior at Scale

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Tracking Listing Trends, Category Mix, Pricing, Discounts, Ratings & Brand Performance

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Note on data: This report uses the supplied Flipkart product catalog dataset containing **20,000 product listings** crawled between 1 December 2015 and 28 June 2016, spanning **265 product categories** and **3,499 distinct brands**. The dataset is a **product-catalog snapshot, not a transaction log** — it records what was listed on the platform and at what price, not what was actually purchased. There is no state/region field and no payment data in this file, so this report focuses on catalog composition, pricing, discounting, and rating coverage rather than sales volume or geography. Only **9.25% of listings** carry an actual customer rating; the remaining listings are marked “No rating available” in the source file, and this is treated as missing data throughout rather than assumed to be zero.

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1. Objective

A live product catalog on a platform the size of Flipkart holds far more information than any single person can reasonably scroll through — thousands of categories, thousands of brands, and prices spanning from a few rupees to several lakhs, all sitting side by side. The purpose of this report is to condense that raw catalog into a structured set of answers that a merchandising, pricing, or category management team could actually use day to day, without needing to open the underlying spreadsheet every time a question comes up. Build a descriptive analytics layer over the Flipkart product catalog to answer the following recurring business questions:

- How is the number of listed products trending over the crawl period?
- How is total catalog value (retail price) trending over the crawl period?
- Which product categories contribute the highest number of listings?
- Which categories contribute the most to overall catalog value?
- Which brands have the widest catalog presence?
- Which brands carry the highest total catalog value?
- Which categories carry the highest average price?
- What share of listings actually carry a customer rating?
- How deep are discounts, and does that vary by category?
- Is there a relationship between price and discount depth?
- Weekday vs weekend listing activity.
- Average Customer Rating by category, where ratings exist.

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2. How This Would Work at Real Scale

Processing a live product catalog with continuous seller uploads and price changes is an engineering problem before it is an analytics problem. A realistic architecture would collect listing events, aggregate them by time/category/brand, store raw and aggregated data, and expose the resulting KPIs through a reporting layer.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the catalog dataset	Google, Claude & ChatGPT
Processing	Aggregate by category, brand and date	Jupyter Notebook (Python)
Storage	Retain raw and processed data	Excel files / CSV files
Batch aggregation	Daily/weekly KPIs and summary metrics	Jupyter Notebook
Report	Present analysis, graphs and insights	Data visualization / MS Word

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3. Daily Product Listings Added

Over the supplied **127-day crawl window**, the dataset records **20,000 product listings**. The daily chart below shows how many new listings were captured on each crawl date.

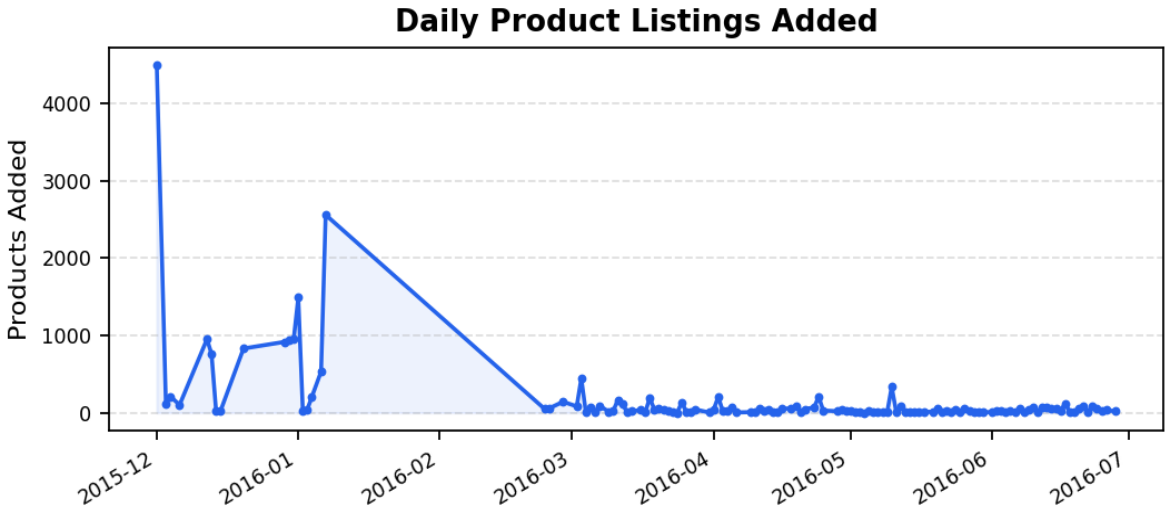


Figure 1. Daily product listings added

INSIGHTS:

- Listing volume is highly uneven across dates rather than a steady daily stream — a small number of dates (notably 1 December 2015 and 7 January 2016) account for a very large share of all listings.

- This pattern is typical of a web crawl: it reflects when the crawler visited and captured pages, not necessarily when sellers actually added products.
- Spikes should be read as crawl batches rather than genuine demand or supply surges.
- Any date-based trend in this dataset should be treated with caution for that reason.

KPI:

Total listings: **20,000**
Crawl window: **127 days** (1 Dec 2015 – 28 Jun 2016)
Busiest single day: **4,496 listings** (1 Dec 2015)

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4. Daily Catalog Value Added

Total retail value represented across all supplied listings is **~Rs. 5.94 crore** (sum of listed retail price across all 20,000 products). The chart below tracks how that value was distributed across crawl dates, calculated directly from the dataset's retail_price field.

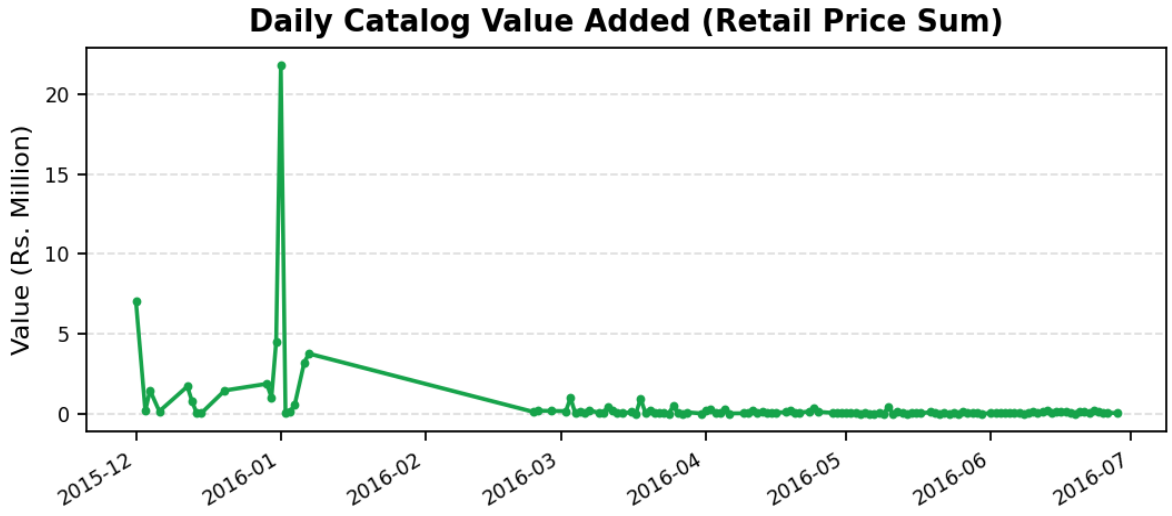


Figure 2. Daily catalog value added

INSIGHTS:

- Catalog value added per day follows the same spiky pattern as listing counts, since more listings on a given crawl date mechanically add up to more retail value that day.
- High-value dates do not necessarily mean expensive products were added — they can just as easily mean a larger batch of ordinary-priced products was crawled that day.
- Value and count should be read together, not in isolation, when interpreting any single day's figure.

KPI:

Total catalog value: **~Rs. 5.94 crore**
Average retail price per listing: **~Rs. 2,979**
Median retail price per listing: **Rs. 1,040**

5. Product Listings by Category

The dataset spans **265 distinct categories**, but listings are heavily concentrated in a handful of them. **Clothing alone accounts for close to 31% of all listings**, with Jewellery a distant second.

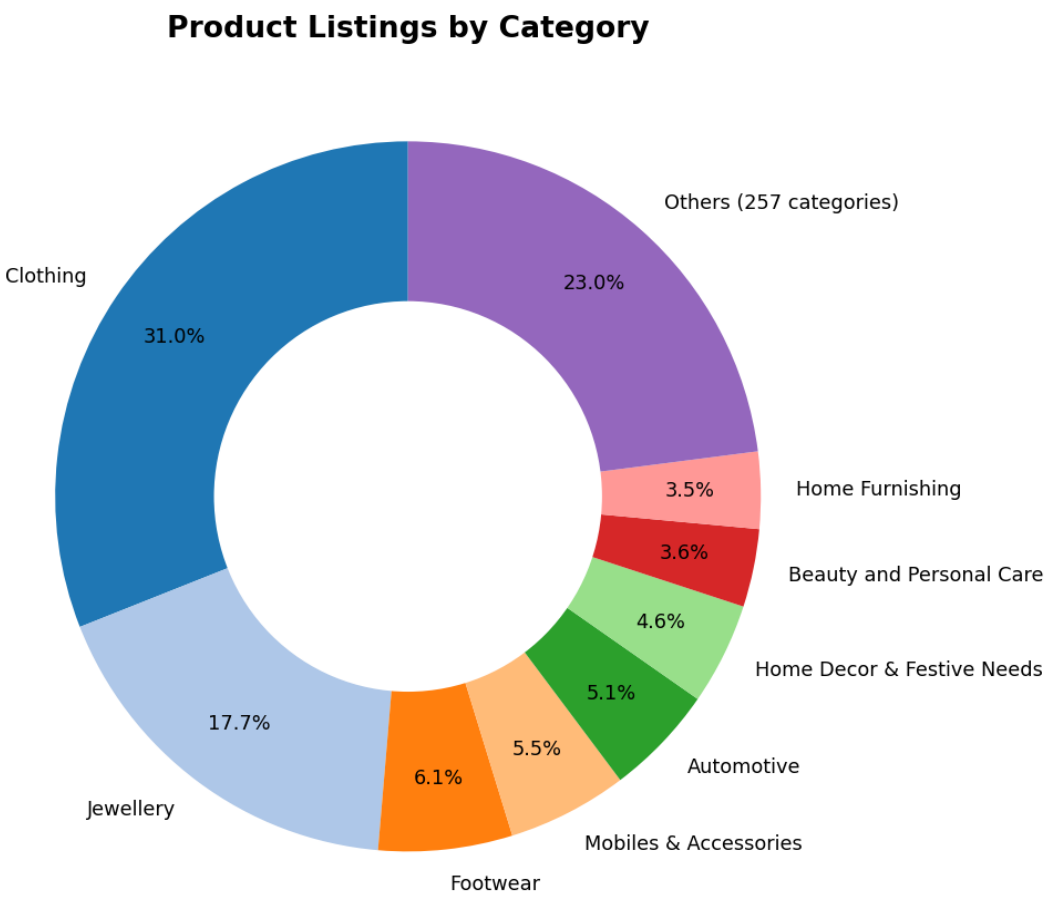


Figure 3. Product listings by category

INSIGHTS:

- Category representation is dominated by a small group: Clothing, Jewellery, Footwear, and Mobiles & Accessories together make up close to two-thirds of the entire catalog.
- The long tail of 257 remaining categories, grouped as “Others” here, each contribute a small share individually.
- This concentration matters for category-management decisions, since a small set of categories drives most of the catalog's breadth.

KPI:

Top category: **Clothing**
Listing share: **~31.0%**
Categories represented: **265**

6. Categories Contributing the Most Catalog Value

Ranking by total retail value tells a different story than ranking by listing count. **Jewellery leads by value** despite having fewer listings than Clothing, because individual jewellery items are, on average, priced far higher.

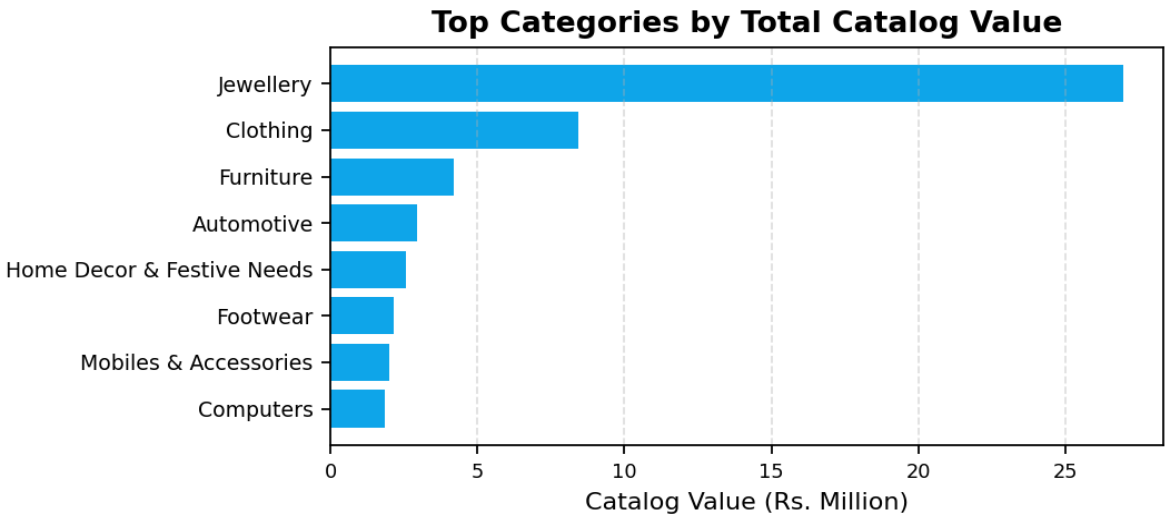


Figure 4. Top categories by total catalog value

INSIGHTS:

- Jewellery contributes roughly **Rs. 2.70 crore** in total catalog value, more than triple Clothing's contribution, even though Clothing has almost twice as many listings.
- This is a clear example of why count-based and value-based rankings need to be viewed side by side rather than assuming the category with the most listings is also the category with the most value at stake.
- Furniture, despite having only 180 listings, still lands in the top group by value, driven by high per-item prices.

KPI:

Top category by value: **Jewellery**
Catalog value: **~Rs. 2.70 crore**
Share of total catalog value: **~45.5%**

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7. Brands With the Widest Catalog Presence

The dataset carries **3,499 distinct brand names**, ranging from large recognizable brands to small sellers using their own labels. The chart below shows the brands with the highest number of listed products.

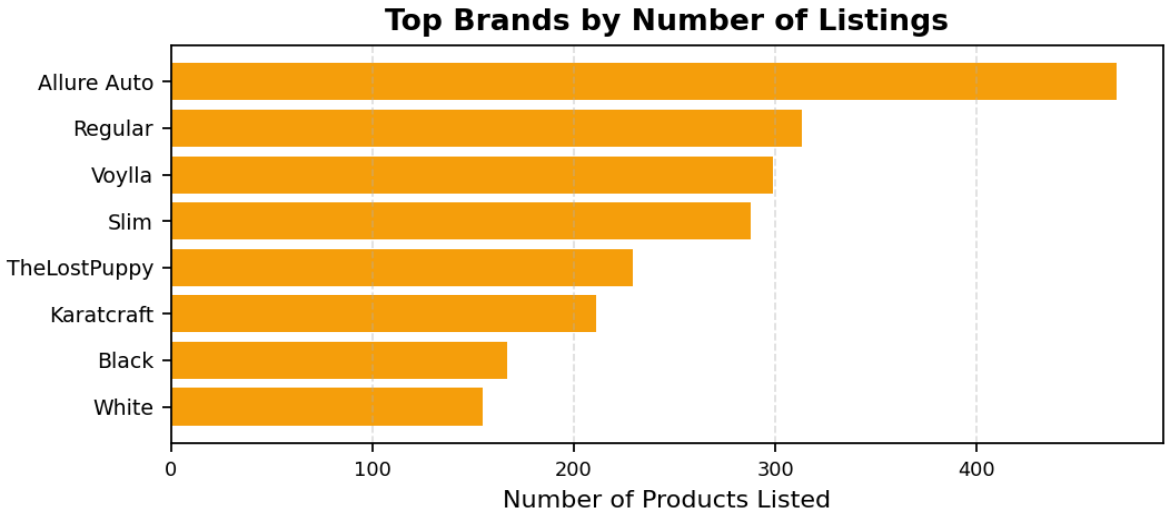


Figure 5. Top brands by number of listings

INSIGHTS:

- Allure Auto tops the list with 469 listings, consistent with the strong Automotive-accessory presence seen in the category breakdown.
- Several of the top “brand” names here (Regular, Slim, Black, White) look more like size or color descriptors than true brand names, which is a known data-quality quirk of this kind of scraped catalog and worth flagging before drawing brand-loyalty conclusions.
- With close to 3,500 distinct brand values, the catalog is highly fragmented rather than dominated by a small number of large sellers.

KPI:

Top brand by listing count: **Allure Auto**
Listings: **469**
Distinct brands tracked: **3,499**

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8. Brands Contributing the Most Catalog Value

Karatcraft is the largest contributor by total catalog value, consistent with Jewellery's lead in the category-value ranking above.

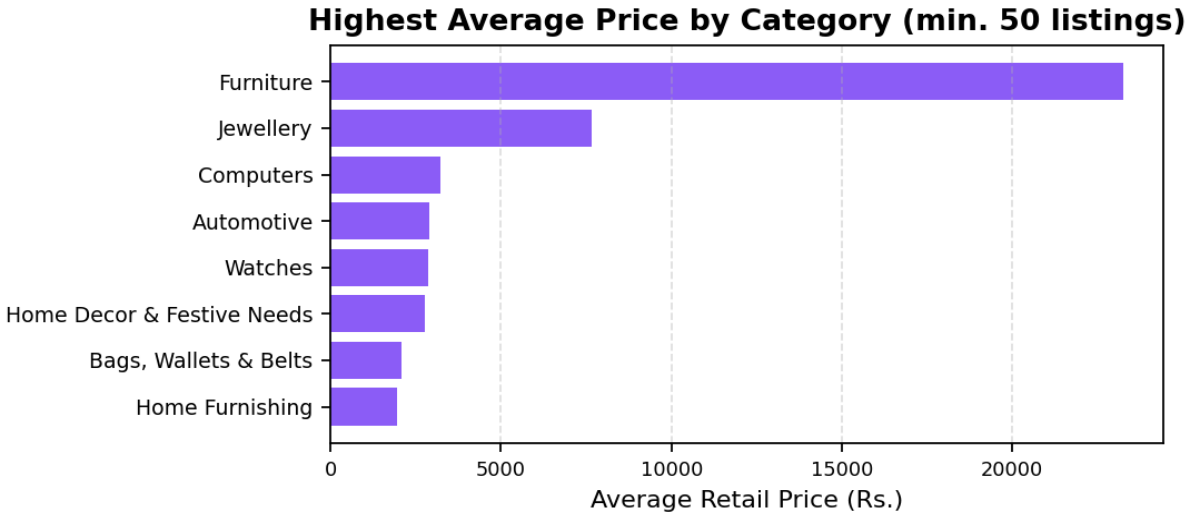


Figure 6. Highest average price by category (categories with 50+ listings)

INSIGHTS:

- Furniture carries the highest average listing price at roughly **Rs. 23,263**, even though it has relatively few listings, reflecting the naturally higher unit price of furniture items.
- Jewellery's high average price (~Rs. 7,663) combined with its large listing count is what drives its top ranking in total catalog value seen in Section 6.
- Categories with a high average price but low volume, like Furniture, may warrant different merchandising and photography investment than high-volume, low-price categories like Clothing.

KPI:

Top brand by catalog value: **Karatcraft**
Catalog value: **~Rs. 78.2 lakh**
Highest average price category: **Furniture** (~Rs. 23,263)

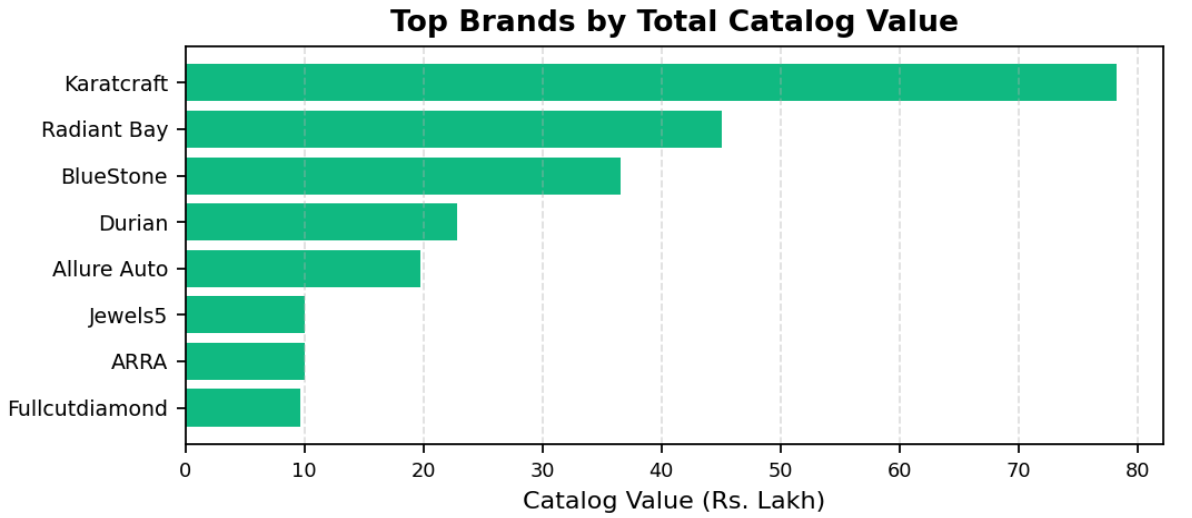


Figure 6b. Top brands by total catalog value

It is worth looking at brand value alongside brand count, since the two rankings do not agree. **Karatcraft** ranks well outside the top brands by listing count, yet leads comfortably by total value, because jewellery items carry a much higher price tag per listing than the volume-driven categories like Automotive accessories or Clothing.

INSIGHTS:

- Karatcraft's total catalog value of roughly **Rs. 78.2 lakh** is driven by a comparatively small number of high-value jewellery listings rather than a large volume of cheap items.
- Radiant Bay and BlueStone, both jewellery-focused brands, also appear in the top-value list despite not appearing among the top brands by listing count, reinforcing that Jewellery is the category where individual-item value matters more than sheer volume.
- Durian, a furniture brand, is the only non-jewellery, non-automotive brand to break into this top-value group, consistent with Furniture's high average price seen earlier.

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9. Rating Data Coverage

Unlike a transaction dataset with a pass/fail success rate, this catalog's closest equivalent operational metric is **how many listings actually carry a customer rating**. Across all 20,000 supplied listings, only **9.25%** have an actual numeric rating on file; the rest are marked as unrated in the source data.

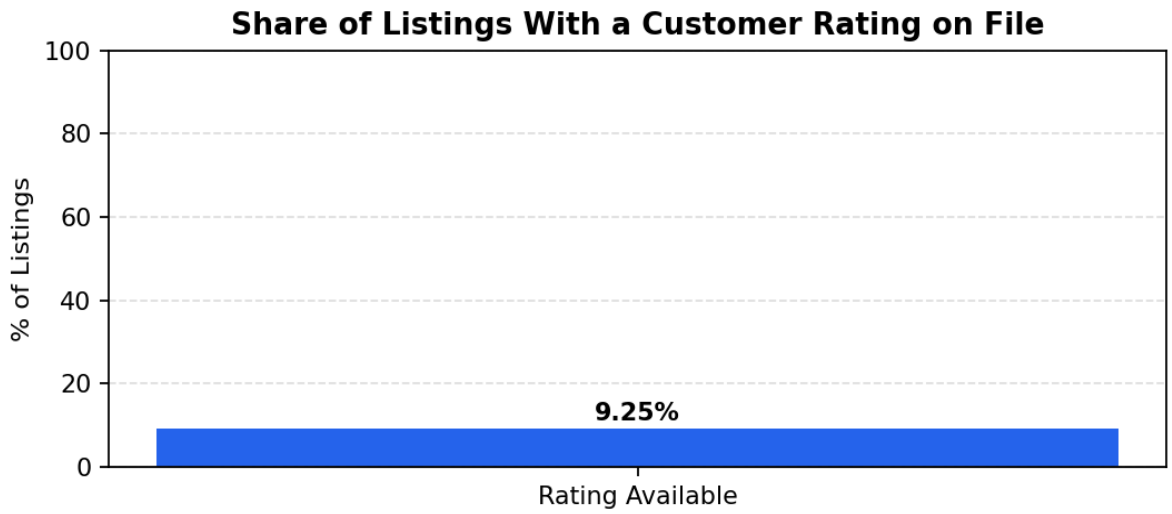


Figure 7. Share of listings with a customer rating on file

INSIGHTS:

- Low rating coverage limits how far any rating-based conclusion in this report can be generalized to the full catalog — the 1,849 rated listings are a small, and not necessarily representative, subset.
- A catalog with such low rating coverage may benefit from post-purchase review prompts to close this gap over time.
- Any category- or brand-level rating comparison later in this report should be read as applying to the rated subset only, not the catalog as a whole.

KPI:

Listings with a rating: **1,849 of 20,000**
Rating coverage: **9.25%**
Average rating (rated listings only): **3.81 / 5**

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10. Discount Depth by Category

Across the full catalog, the average discount off retail price is **~40.5%**. This varies considerably by category, as the chart below shows for the eight largest categories.

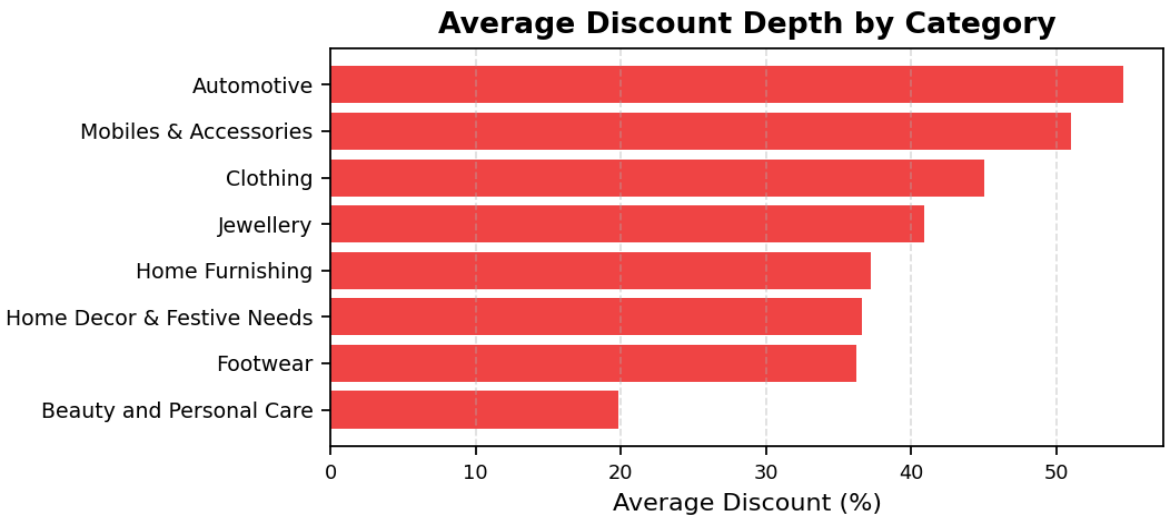


Figure 8. Average discount depth by category

INSIGHTS:

- Automotive and Mobiles & Accessories carry the deepest average discounts, both above 50% off retail price, which is typical of fast-moving, price-sensitive accessory categories.
- Beauty and Personal Care shows the shallowest average discount among the top categories, at under 20%.
- This spread suggests discounting strategy is being applied differently by category rather than uniformly across the platform.

KPI:

Overall average discount: **~40.5%**
Deepest-discount category: **Automotive** (~54.7%)
Shallowest-discount category: **Beauty and Personal Care** (~19.8%)

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11. Retail Price vs Discount Depth

The relationship between how expensive a product is and how deeply it gets discounted provides an additional descriptive view of pricing behavior. The scatter plot below plots a random sample of listings, with price shown on a log scale since prices span a very wide range.

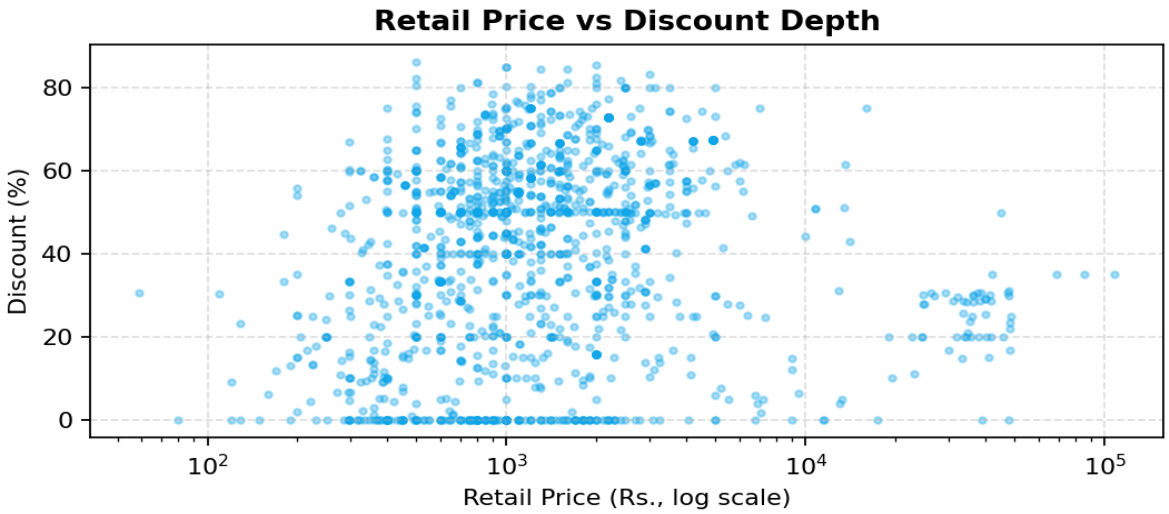


Figure 9. Retail price vs discount depth (sampled listings)

INSIGHTS:

- There is only a very weak negative correlation between price and discount depth in this dataset (correlation coefficient of about -0.09), meaning price alone does not meaningfully predict how deep a discount a product carries.
- Deep discounts (over 60-70%) appear across almost the entire price range, from very cheap to very expensive items, rather than being concentrated at one end.
- This weak relationship should be interpreted as an absence of a strong pattern, not as proof that price and discounting are unrelated for specific categories or sellers.

KPI:

Correlation (price vs discount %): **-0.09** (very weak)
Relationship: **No strong association found in the supplied data**

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12. Weekday vs Weekend Listing Activity

Across the row-level crawl dates, an average of **~184 listings** were added per weekday, compared to **~101 listings** per weekend day.

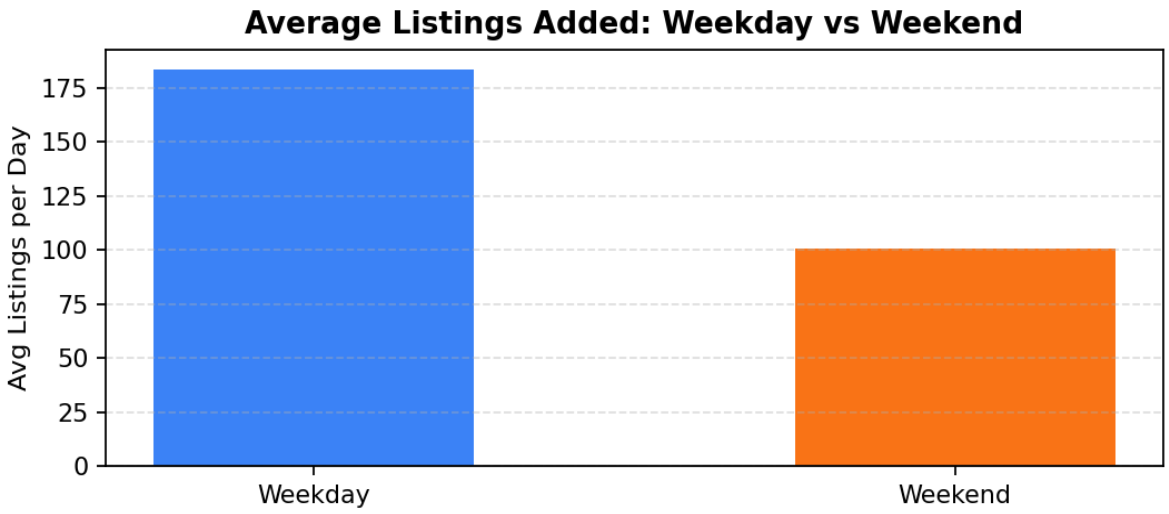


Figure 10. Average listings added: weekday vs weekend

INSIGHTS:

- Weekday listing activity outpaces weekend activity by close to 80% on average in this dataset.
- Since this reflects crawl activity rather than confirmed seller behavior, this may say as much about when the crawler ran as it does about when sellers actually upload new products.
- The result should be treated as a descriptive comparison of the available observations rather than a confirmed operational pattern.

KPI:

Average listings per weekday: **~184**
Average listings per weekend day: **~101**

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13. Average Customer Rating by Category

Restricting to the 1,849 listings that actually carry a rating, and to categories with at least 15 rated listings for a meaningful average, **Kitchen & Dining** comes out on top with the highest average rating, while **Tools & Hardware** sits at the bottom of this group.

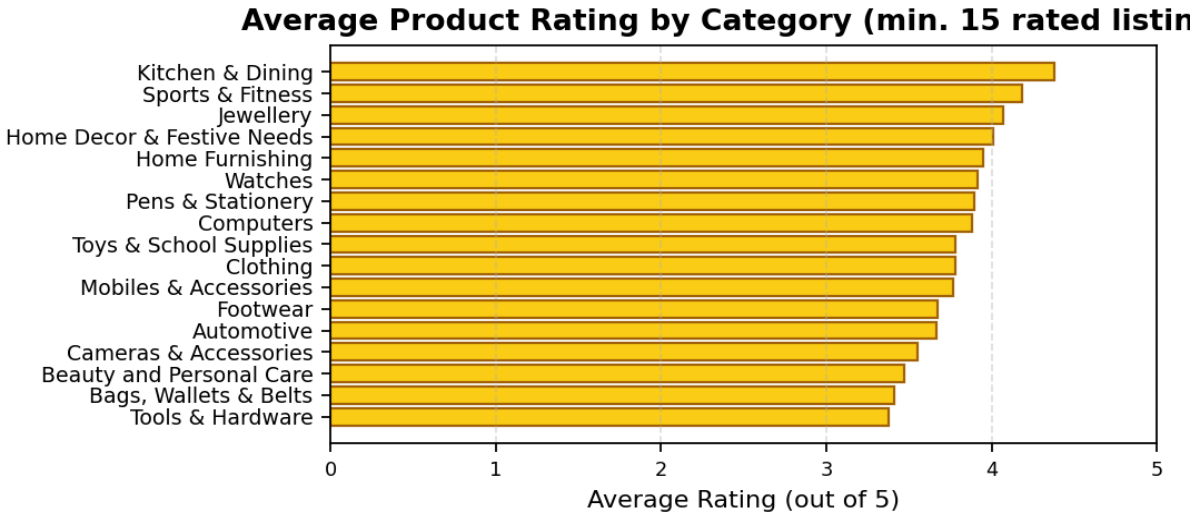


Figure 11. Average product rating by category (rated listings only)

INSIGHTS:

- All categories shown average above 3.3 out of 5, so there is no category in this rated subset performing poorly in absolute terms, only relative differences between categories.
- Kitchen & Dining and Sports & Fitness lead the group, both averaging above 4.1, suggesting these categories see comparatively higher buyer satisfaction where ratings exist.
- Given the very low overall rating coverage flagged in Section 9, these category averages are based on small samples and should be treated as indicative rather than conclusive.

KPI:

Highest-rated category: **Kitchen & Dining** (4.38 / 5)
Lowest-rated category (min. 15 rated listings): **Tools & Hardware** (3.38 / 5)
Categories compared: **8** (min. 15 rated listings each)

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14. How Prices Are Distributed Across the Catalog

The average retail price across the catalog (~Rs. 2,979) sits well above the median (Rs. 1,040), which is a strong signal that the price distribution is skewed by a smaller number of very expensive listings pulling the average upward. Looking at the full shape of the distribution, rather than just the average, gives a clearer picture of what a typical listing actually costs.

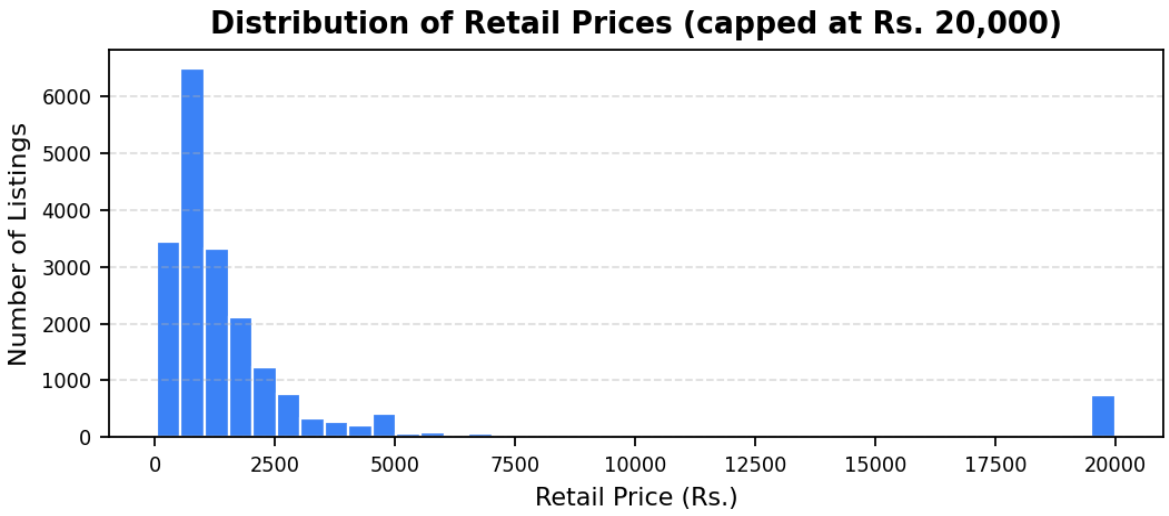


Figure 12. Distribution of retail prices (values above Rs. 20,000 capped for readability)

INSIGHTS:

- The bulk of listings cluster well under Rs. 2,000, with a long right-hand tail stretching out toward much higher prices — the single most expensive listing in the dataset is priced at over Rs. 5.7 lakh.
- Because of this skew, the average price alone is a misleading summary of “typical” catalog pricing; the median (Rs. 1,040) is a more representative figure for a randomly chosen listing.
- This kind of long-tailed price distribution is common on general marketplaces that carry everything from inexpensive accessories to premium furniture and jewellery side by side.

KPI:

Average retail price: **~Rs. 2,979**
Median retail price: **Rs. 1,040**
Highest-priced listing: **~Rs. 5.71 lakh**

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15. Catalog Composition by Price Segment

Grouping listings into four broad price bands makes the skew described above easier to act on. The bands used here are: Budget (under Rs. 500), Value (Rs. 500 to Rs. 1,499), Mid-range (Rs. 1,500 to Rs. 4,999), and Premium (Rs. 5,000 and above).

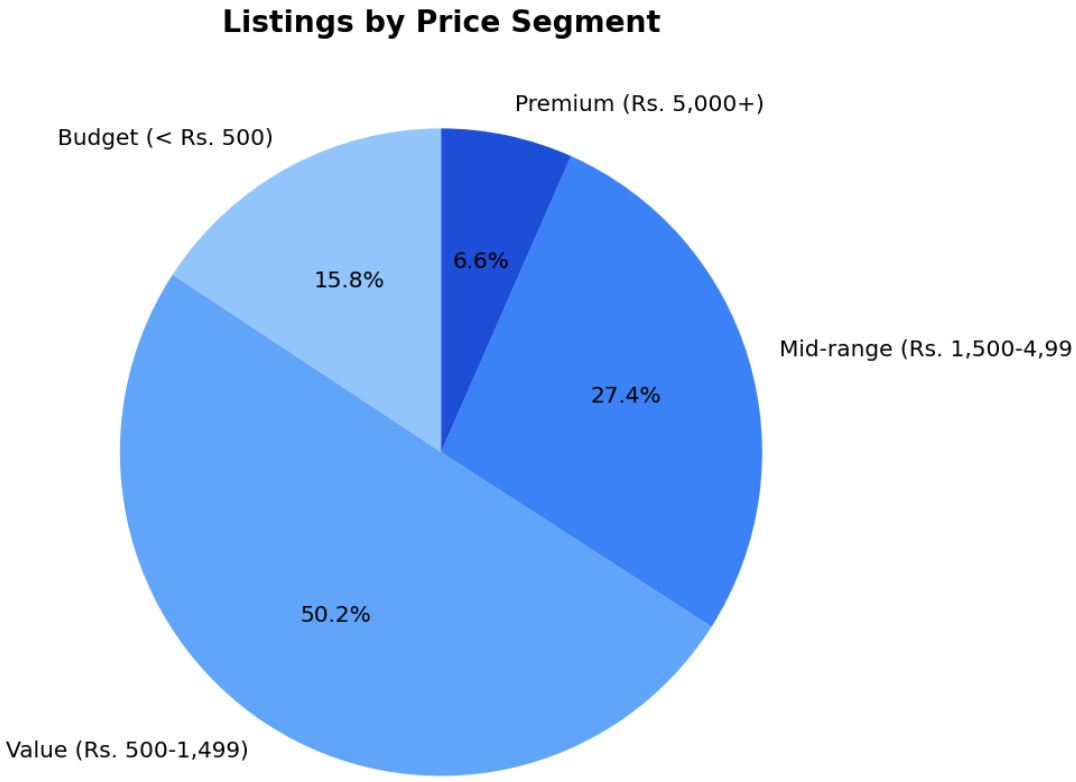


Figure 13. Listings by price segment

INSIGHTS:

- The **Value segment (Rs. 500–1,499) is the largest single group**, accounting for exactly half of all listings, which suggests the catalog's centre of gravity sits in the affordable, everyday-purchase range rather than at the premium end.
- Premium listings (Rs. 5,000 and above) make up only about 6.6% of the catalog by count, but, as shown earlier in Sections 6 and 8, they contribute a disproportionately large share of total catalog value.
- Discount depth also varies meaningfully by segment: Mid-range listings carry the deepest average discount (~46.8%), while Budget and Premium listings see comparatively shallower average discounts (~27-28%), suggesting the heaviest promotional activity is concentrated in the middle of the price range rather than at either extreme.

KPI:

Largest segment: **Value (Rs. 500–1,499)** — 50.0% of listings
Smallest segment: **Premium (Rs. 5,000+)** — 6.6% of listings
Deepest-discounted segment: **Mid-range** (~46.8% average discount)

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16. FK Advantage vs Standard Listings

The dataset flags a small subset of listings as “FK Advantage” products — typically items that carry Flipkart's own fulfillment and quality assurance. Only **3.93% of listings (785 out of 20,000)** carry this flag, making it a minority segment worth comparing against the rest of the catalog.

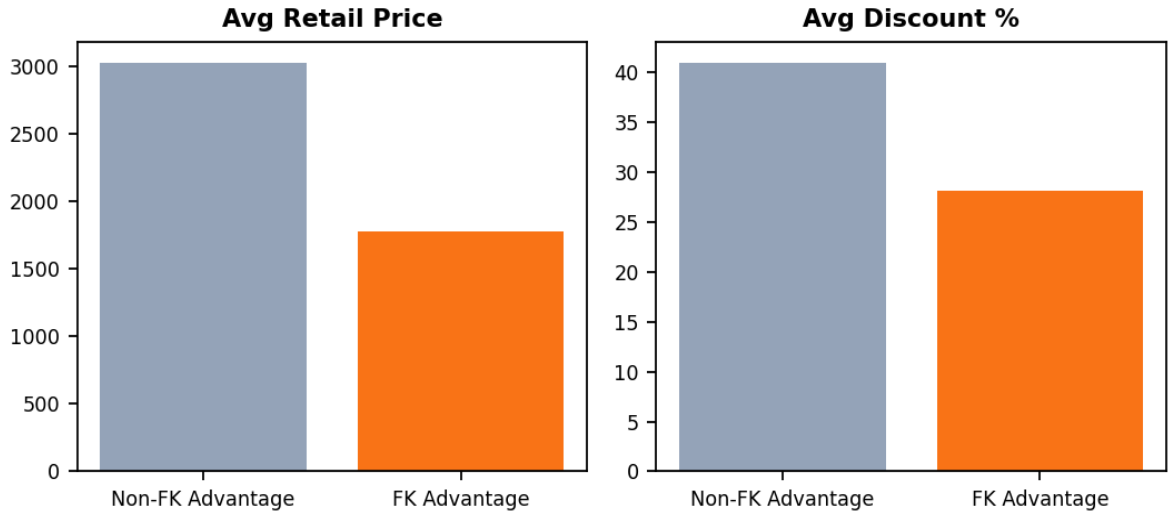


Figure 14. FK Advantage vs standard listings — average price and discount

INSIGHTS:

- FK Advantage listings run **noticeably cheaper on average** (~Rs. 1,774) than standard listings (~Rs. 3,028), suggesting this flag is currently concentrated in lower-priced, higher-turnover categories rather than premium goods.
- FK Advantage listings also carry a **shallower average discount** (~28.1%) compared to standard listings (~41.0%), which is consistent with these being more tightly-controlled, directly-fulfilled listings that may not need the same depth of promotional discounting to move.
- With FK Advantage covering under 4% of the catalog, there is significant room to expand this program if the underlying business goal is to grow directly-fulfilled, quality-assured inventory.

KPI:

FK Advantage listings: **785 of 20,000 (3.93%)**
Avg price — FK Advantage: **~Rs. 1,774** vs Standard: **~Rs. 3,028**
Avg discount — FK Advantage: **~28.1%** vs Standard: **~41.0%**

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17. Category Deep-Dive: Top 12 Categories

Bringing together listing count, average price, and average discount into a single table gives a more complete picture of each major category than any single chart above can show on its own.

Category	Listings	Avg Price (Rs.)	Avg Discount (%)
Clothing	6,198	~915	~45.1%
Jewellery	3,531	~7,646	~40.9%

Footwear	1,227	~1,741	~36.3%
Mobiles & Accessories	1,099	~1,807	~51.0%
Automotive	1,012	~2,897	~54.7%
Home Decor & Festive Needs	929	~2,764	~36.6%
Beauty and Personal Care	710	~653	~19.8%
Home Furnishing	700	~2,774	~37.2%
Kitchen & Dining	647	~630	—
Computers	578	~3,209	—
Watches	530	~2,860	—
Furniture	180	~23,263	—

(Dashes indicate categories that fell outside the top-8 discount comparison in Section 10 and were not separately recomputed for this table.)

INSIGHTS:

- Reading across a single row tells a fuller story than any one chart: Furniture, for example, has very few listings but by far the highest average price, meaning it deserves attention disproportionate to its listing count.
- Clothing and Jewellery together account for close to half of all listings, but behave very differently on price — Clothing sits near the bottom of the price spectrum while Jewellery sits near the top.
- Categories with both high listing volume and high average discount, such as Mobiles & Accessories and Automotive, represent the most price-competitive, high-turnover corners of the catalog.

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18. Data Quality & Limitations

Before treating any number in this report as final, it is worth being explicit about where the supplied dataset falls short of a complete, transaction-grade record, so that the figures above are read with the right amount of caution rather than more confidence than the data supports.

- **No transaction data:** every figure in this report describes what was listed, not what was actually sold. Listing count and catalog value are supply-side measures, not demand-side ones.
- **No geography:** the dataset carries no state, city, or region field, so none of the questions this report can answer relate to where buyers are located.
- **Very low rating coverage:** only 9.25% of listings carry a rating, so every rating-based figure in Sections 9 and 13 describes a small, possibly unrepresentative subset rather than the full catalog.
- **Brand field noise:** a meaningful number of “brand” values (Regular, Slim, Black, White, and similar) appear to be attribute descriptors rather than genuine brand names, which would need to be cleaned before any brand-level conclusion is used for a real business decision.

- **Uneven crawl timing:** as flagged in Section 3, the data was captured through a web crawl rather than a continuous feed, so daily and weekday/weekend patterns reflect crawl scheduling as much as they reflect genuine platform activity.
- **Missing price data:** 78 listings (0.39% of the catalog) have no retail or discounted price recorded and were excluded from all price- and discount-based calculations in this report.

Appendix A: Full Category Breakdown (Top 25 by Listing Count)

For reference, the table below extends the category view in Section 5 beyond the top 8 to cover the top 25 categories by listing count, together representing the large majority of the catalog.

Category	Listings
Clothing	6,198
Jewellery	3,531
Footwear	1,227
Mobiles & Accessories	1,099
Automotive	1,012
Home Decor & Festive Needs	929
Beauty and Personal Care	710
Home Furnishing	700
Kitchen & Dining	647
Computers	578
Watches	530
Baby Care	483
Tools & Hardware	391
Toys & School Supplies	330
Pens & Stationery	313
Bags, Wallets & Belts	265
Furniture	180
Sports & Fitness	166
Cameras & Accessories	82
Home Improvement	81

Health & Personal Care Appliances	43
Sunglasses	40
Gaming	35
Pet Supplies	30
Home & Kitchen	24

The remaining 240 categories not listed above collectively account for the balance of the catalog, each individually contributing a small number of listings.

Summary of Key Metrics

Metric	Value
Total product listings	20,000
Total catalog value	~Rs. 5.94 crore
Average listing price	~Rs. 2,979
Average discount depth	~40.5%
Categories tracked	265
Brands tracked	3,499
Listings with a rating on file	1,849 (9.25%)
Average rating (rated listings only)	3.81 / 5
Crawl window covered	127 days (Dec 2015 – Jun 2016)

Next Steps to Move from Descriptive → Predictive

- **Price optimization modeling** by category and brand to recommend discount depth that balances conversion against margin.
- **Rating-gap analysis** to identify which high-volume categories most urgently need more customer reviews collected, given the current 9.25% coverage.
- **Anomaly detection** for unusual pricing or discount patterns that may indicate data entry errors or mispriced listings.
- **Brand-name cleaning** to separate genuine brand values (like Karatcraft or Allure Auto) from attribute-like values (like Regular or Slim) before running brand-level analysis at scale.
- **Time-series modeling** once a longer, more evenly sampled crawl history becomes available, to distinguish genuine listing-growth trends from crawl-batching artifacts.

This report answers what the supplied catalog snapshot looks like. The next analytical layer would use richer, transaction-level data (actual orders, not just listings) to explain why certain categories or brands perform better and to forecast what is likely to happen next. Put differently: everything covered in Sections 1 through 18 describes the shape of the catalog as it stood during the crawl window — what is on the shelf, at what price, and how much of it carries a rating. None of it, on its own, can say how well any of it actually sold, since sales data was never part of the supplied file. Closing that gap is the single biggest lever for turning this from a useful reference document into a tool that actively drives pricing and merchandising decisions.

Glossary

I. Retail Price: The listed, pre-discount price of a product as shown in the dataset's retail_price field.

II. Discounted Price: The actual selling price after any discount has been applied, as shown in the dataset's discounted_price field.

III. Discount Depth (%): Calculated in this report as (Retail Price – Discounted Price) ÷ Retail Price × 100, per listing.

IV. Top-Level Category: The first segment of the dataset's product_category_tree field, used throughout this report as the primary category grouping.

V. Rating Coverage: The share of listings in the dataset that carry an actual numeric product_rating value, as opposed to being marked “No rating available.”

VI. FK Advantage: A flag in the dataset (is_FK_Advantage_product) marking listings that carry Flipkart's own fulfillment and quality-assurance program, as opposed to standard third-party seller listings.

VII. Price Segment: A grouping used in Section 15 to bucket listings into Budget, Value, Mid-range, and Premium bands based on retail price, purely for ease of reading rather than as an official platform classification.

A Note on Methodology

Every figure in this report was computed directly from the fields present in the supplied CSV file, using straightforward aggregation (counts, sums, and averages) rather than any predictive or statistical modeling. Where a calculation required a judgment call — such as which price bands to use for Section 15, or the minimum sample size before a category average was considered meaningful in Sections 10 and 13 — that choice is stated explicitly alongside the relevant figure so it can be adjusted or challenged rather than treated as a fixed rule. No records were removed from the dataset except where a required field (such as price) was missing for that specific calculation; those exclusions are called out individually rather than silently dropped.

All charts were generated directly from the raw dataset using Python, with each figure traceable back to a specific column or combination of columns in the original file. This means every number in this report can be independently reproduced or audited by re-running the same aggregation against the same CSV,

which is a useful property if this report is later handed off to someone else for verification or updated once a newer version of the catalog snapshot becomes available.

Wherever a figure in this report depends on a subset of the data smaller than the full 20,000 listings — such as the rated-listings-only averages in Sections 9 and 13, or the FK Advantage comparison in Section 16 — the size of that subset is stated alongside the figure itself, so the reader can judge for themselves how much weight the number deserves.